

5(a)(i)	Since copper has one conduction electron per atom, the number density of charge carriers is equal to the number of atoms per unit volume. $\text{number of mole} = \frac{\text{mass}}{\text{molar mass}}$ number of mole per unit volume $= \frac{\text{density}}{\text{molar mass}}$ $= \frac{8.9 \times 10^3}{63.5 \times 10^{-3}}$ $= 1.40157 \times 10^5 \text{ m}^{-3}$	[1]

No.	Solution	Remarks
	number of atoms per unit volume $= \text{number of mole per unit volume} \times N_A$ $= 1.40157 \times 10^5 \times 6.02 \times 10^{23}$ $= 8.44 \times 10^{28}$ $\approx 8.4 \times 10^{28} \text{ m}^{-3}$ Hence number density of charge carriers $= 8.4 \times 10^{28} \text{ m}^{-3}$	[1] answer before rounding off
5(a)(ii)	Using $I = Anvq$ $17 \times 10^{-3} = \pi \times \left(\frac{1.8 \times 10^{-3}}{2} \right)^2 \times 8.4 \times 10^{28} \times v \times 1.60 \times 10^{-19}$ $v = 4.97 \times 10^{-7}$ $\approx 5.0 \times 10^{-7} \text{ m s}^{-1}$	[1] for correct substitution [1] for correct answer
5(b)	<u>Current is the same</u> in all segments of the wire, including the thinner segment. For the thinner segment, the <u>cross-sectional area is smaller</u> . The charge carriers hence move with a <u>higher drift velocity</u> (number density of charge carriers does not change).	[1] same current and smaller cross-sectional area [1] higher drift velocity
6(a)	Faraday's law of electromagnetic induction states that the	[1]